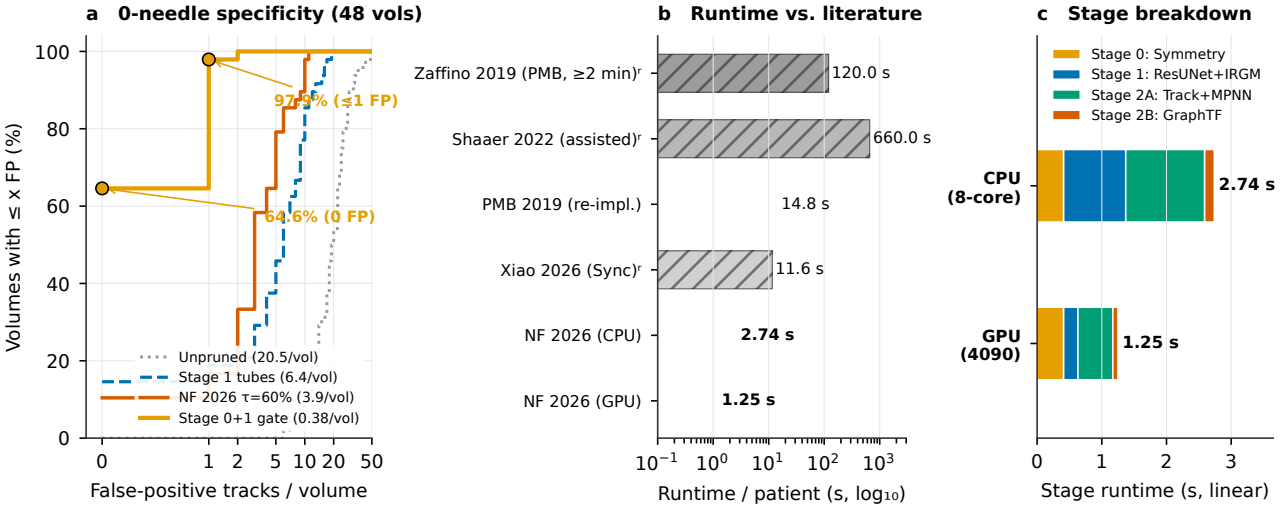
a 0-needle specificity (48 vols)
Volumes with ≤ x FP (%)
False-positive tracks / volume
97.9% (≤1 FP)
64.6% (0 FP)
Unpruned (20.5/vol)
Stage 1 tubes (6.4/vol)
NF 2026 τ=60% (3.9/vol)
Stage 0+1 gate (0.38/vol)
b Runtime vs. literature
Zaffino 2019 (PMB, ≥2 min)r
Shaaer 2022 (assisted)r
PMB 2019 (re-impl.)
Xiao 2026 (Sync)r
NF 2026 (CPU)
NF 2026 (GPU)
120.0 s
660.0 s
14.8 s
11.6 s
2.74 s
1.25 s
Runtime / patient (s, log10)
c Stage breakdown
Stage 0: Symmetry
Stage 1: ResUNet+IRGM
Stage 2A: Track+MPNN
Stage 2B: GraphTF
CPU (8-core)
GPU (4090)
2.74 s
1.25 s
Stage runtime (s, linear)